Worksheet: MOI Transformation (finding Principals $\mathbf{I}_{MAX}$ and $\mathbf{I}_{MIN}$) using Equations

<table>
<tr><th>Step/Equations</th><th>Inputs/Calculations</th></tr>
<tr><td>Original Area (Shape):</td><td>I_x =

I_y =

I_{xy} =</td></tr>
<tr><td>Intermediate Calculations:
$I_{AVG} = MC\ Center = \left(\frac{I_x + I_y}{2}\right)$
$\left(\frac{I_x - I_y}{2}\right) =$
$Radius\ R = I_{xyMAX} = \sqrt{\left(\frac{I_x - I_y}{2}\right)^2 + I_{xy}^2}$</td><td></td></tr>
<tr><td>Principal MOI's: $I_{MAX}, I_{MIN} = I_{AVG} \mp R$</td><td></td></tr>
<tr><td>Principal Angles: (Also compute 2θ_p…)
$\theta_p = \frac{1}{2} tan^{-1}\left(\frac{-I_{xy}}{\left(\frac{I_x - I_y}{2}\right)}\right)$</td><td></td></tr>
<tr><td>Which I_{MAX}, I_{MIN} is at θ_p?
$I_\theta = \left(\frac{I_x + I_y}{2}\right) + \left(\frac{I_x - I_y}{2}\right) cos(2\theta) - I_{xy} sin(2\theta)$</td><td></td></tr>
<tr><td colspan="2">Final Picture of Shape with I_{MAX}, I_{MIN}, θ_p:</td></tr>
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